Minkowskian open/closed conformal field theory possibly without vacuum: the Cardy case

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Outline

Part I

Geometric Setup

Closed and Open Strings

Closed and open strings

2D CFT studies the propagation and interaction of closed strings

$$S^1 \simeq \mathbb{R}/2\pi\mathbb{Z}$$

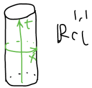
and open strings

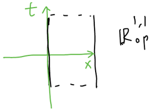
$$[0,\pi].$$

The corresponding Minkowski space-times are

$$\mathbb{R}^{1,1}_{\mathrm{cl}} = S^1 \times \mathbb{R}, \qquad \mathbb{R}^{1,1}_{\mathrm{op}} = [0,\pi] \times \mathbb{R}.$$

$$(x,t) \in \mathbb{R}^{1,1}_{\mathrm{cl}} \text{ or } \mathbb{R}^{1,1}_{\mathrm{op}},$$
$$(ds)^2 = (dt)^2 - (dx)^2,$$
$$u^+ = t + x, \qquad u^- = t - x,$$
$$(ds)^2 = du^+ du^-.$$





Conformal Diffeomorphisms

Let $\mathrm{Diff}_+(S^1)$ be the group of orientation-preserving diffeomorphisms of S^1, and let

$$\mathscr{G} = \widetilde{\mathrm{Diff}_+(S^1)}$$

be its universal cover. Here

$$\mathrm{Cf}^+(\mathbb{R}^{1,1}_{\mathrm{cl}}),\ \mathrm{Cf}^+(\mathbb{R}^{1,1}_{\mathrm{op}})$$

denote orientation-preserving, time-direction-preserving conformal diffeomorphism groups. Elements $F \in \mathscr{G}$ can be viewed as smooth functions $F : \mathbb{R} \to \mathbb{R}$ such that

$$F(x + 2\pi) = F(x) + 2\pi, \qquad F'(x) > 0.$$

In light-cone coordinates, the relevant actions are

$$(g_1, g_2)(u^+, u^-) = (g_1 u^+, g_2 u^-) \quad \text{on } \mathbb{R}^{1,1}_{\mathrm{cl}},$$

$$g(u^+, u^-) = (g u^+, g u^-) \quad \text{on } \mathbb{R}^{1,1}_{\mathrm{op}}.$$

Both actions are well-defined. This allows us to classify $\mathrm{Cf}^+(\mathbb{R}^{1,1}_{\mathrm{cl}})$ and $\mathrm{Cf}^+(\mathbb{R}^{1,1}_{\mathrm{op}})$.

$\mathrm{Cf}^+(\mathbb{R}^{1,1}_{\mathrm{cl}})$

The action of $\mathscr{G} \times \mathscr{G}$ on $\mathbb{R}^{1,1}_{\mathrm{cl}}$ gives the covering exact sequence

$$1 \longrightarrow \mathbb{Z} \longrightarrow \mathscr{G} \times \mathscr{G} \longrightarrow \mathrm{Cf}^+(\mathbb{R}^{1,1}_{\mathrm{cl}}) \longrightarrow 1.$$

Kernel

The subgroup $\mathbb{Z}$ is generated by

$$\big(\rho(2\pi), \rho(-2\pi)\big) \in \mathscr{G} \times \mathscr{G},$$

where

$$\rho : \mathbb{R} \longrightarrow \mathscr{G}$$

is the one-parameter rotation subgroup.

$\mathrm{Cf}^+(\mathbb{R}^{1,1}_{\mathrm{op}})$

Let $\mathscr{G}$ act on $\mathbb{R}^{1,1}_{\mathrm{op}}$ by

$$g(u^+, u^-) = (gu^+, gu^-).$$

Classification

The action is well-defined and gives

$$\mathscr{G} \simeq \mathrm{Cf}^+(\mathbb{R}^{1,1}_{\mathrm{op}}).$$

- Closed CFT uses $\mathscr{G} \times \mathscr{G}$.
- Open CFT uses $\mathscr{G}$.
- With central charge c, we use the central extension $\mathscr{G}_c$.

Part II

Boundary Fields and Open String State Spaces

Adapting Fuchs–Schweigert–Runkel

Adapting Fuchs–Schweigert–Runkel to the operator algebraic approach of CFT, we expect a 2D open/closed CFT with central charge c as follows.

- Boundary conditions are labeled by indices

$$i, j, k, \dots .$$

- The open-string state space with left and right boundary conditions i, j is

$$\mathcal{H}_{\mathrm{op}(i,j)}.$$

- Boundary operators preserving a boundary condition i form a generally non-local net

$$\mathcal{B}_{\mathrm{op}(i)}: \ \widetilde{I} \in \widetilde{\mathcal{J}} \longmapsto \mathcal{B}_{\mathrm{op}(i)}(\widetilde{I}) \subset \mathfrak{L}(\mathcal{H}_{\mathrm{op}(i,i)}).$$

  where $\mathcal{B}_{\mathrm{op}(i)}(\widetilde{I})$ is a von Neumann algebra (i.e. a $*$-subalgebra closed under strong operator topology).

Arg-valued intervals

$$\widetilde{\mathcal{J}} = \{\widetilde{I} : \ \widetilde{I} \text{ is a component over } I \subset S^1\}$$

$\widetilde{\mathcal{J}}$

Let

$$\mathcal{J} = \{I \subset S^1 : \ I \text{ is non-empty, open, non-dense, and connected}\}.$$

Let

$$\widetilde{\mathcal{J}} = \{\widetilde{I} : \ \widetilde{I} \text{ is a component of } I \text{ in } \mathbb{R} \to \mathbb{R}/2\pi\mathbb{Z} = S^1\}.$$

Equivalently,

$$\widetilde{\mathcal{J}} = \{\text{open intervals in } \mathbb{R} \text{ of length } 0 < |\widetilde{I}| < 2\pi\}.$$

Non-local boundary net

For a boundary condition i,

$$\mathcal{B}_{\mathrm{op}(i)} : \widetilde{I} \in \widetilde{\mathcal{J}} \longmapsto \mathcal{B}_{\mathrm{op}(i)}(\widetilde{I}) \subset \mathfrak{L}(\mathcal{H}_{\mathrm{op}(i,i)})$$

where $\mathcal{B}_{\mathrm{op}(i)}(\widetilde{I})$ is a von Neumann algebra.

Conformal Covariance

The boundary net is conformally covariant. Let $\mathscr{G}_c$ be the central extension of $\mathscr{G}$ with central charge c.

Covariance

There is a strongly continuous unitary representation of $\mathscr{G}_c$ on $\mathcal{H}_{\mathrm{op}(i,i)}$, indeed on $\mathcal{H}_{\mathrm{op}(i,j)}$, such that

$$g\,\mathcal{B}_{\mathrm{op}(i)}(\widetilde{I})\,g^{-1} = \mathcal{B}_{\mathrm{op}(i)}(g\widetilde{I}), \qquad \widetilde{I} \in \widetilde{\mathcal{J}}.$$

The same convention applies to the right action of $\mathcal{B}_{\mathrm{op}(i)}$.

Left and Right Actions

On the open-string state space $\mathcal{H}_{\mathrm{op}(i,j)}$, we have a left action of $\mathcal{B}_{\mathrm{op}(i)}$ and a right action of $\mathcal{B}_{\mathrm{op}(j)}$:

$$\mathcal{B}_{\mathrm{op}(i)}(\widetilde{I}) \curvearrowright \mathcal{H}_{\mathrm{op}(i,j)} \curvearrowleft \mathcal{B}_{\mathrm{op}(j)}(\widetilde{J}),$$

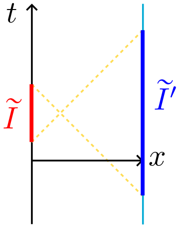
where the first one is the left action and the second one is the right action.

Boundary-boundary locality

If $\widetilde{J}$ is clockwise to $\widetilde{I}$, i.e. $0 < \arg z - \arg \zeta < 2\pi$ for each $z \in \widetilde{I}$ and $\zeta \in \widetilde{J}$, then

$$\big[\mathcal{B}_{\mathrm{op}(i)}(\widetilde{I}), \mathcal{B}_{\mathrm{op}(j)}(\widetilde{J})\big] = 0.$$

Boundary-Boundary Haag Duality



For $\mathcal{H}_{\mathrm{op}(i,j)}$, the nets $\mathcal{B}_{\mathrm{op}(i)}$ and $\mathcal{B}_{\mathrm{op}(j)}$ act on the left and right boundaries.

Boundary-boundary Haag duality

The following von Neumann algebras are commutants of each other:

$$\mathcal{B}_{\mathrm{op}(i)}(\widetilde{I}), \qquad \mathcal{B}_{\mathrm{op}(j)}(\widetilde{I}').$$

This is called the **boundary-boundary Haag duality**.

Here, $\widetilde{I}'$ is the largest element in $\widetilde{\mathcal{J}}$ clockwise to $\widetilde{I}$. In particular, if $\widetilde{J}$ is clockwise to $\widetilde{I}$, then

$$\big[\mathcal{B}_{\mathrm{op}(i)}(\widetilde{I}), \mathcal{B}_{\mathrm{op}(j)}(\widetilde{J})\big] = 0.$$

Morita Equivalent Boundary Conditions

That $\mathcal{B}_{\mathrm{op}(i)}$ and $\mathcal{B}_{\mathrm{op}(j)}$ are commutants of each other on $\mathcal{H}_{\mathrm{op}(i,j)}$ implies that $\mathcal{B}_{\mathrm{op}(i)}$ and $\mathcal{B}_{\mathrm{op}(j)}$ are **Morita equivalent**.
This would imply that they produce the same closed CFT. This still needs to be proved in the operator algebra framework.

Conjugation

$$\overline{\mathcal{H}_{\mathrm{op}(i,j)}} = \mathcal{H}_{\mathrm{op}(j,i)}.$$

Composition of open strings

$$\mathcal{H}_{\mathrm{op}(i,j)} \boxtimes_{\mathcal{B}_{\mathrm{op}(j)}} \mathcal{H}_{\mathrm{op}(j,k)} \simeq \mathcal{H}_{\mathrm{op}(i,k)}.$$

Here $\boxtimes$ denotes the Connes fusion.

Cardy-Type CFT

We say that the CFT is of Cardy type if there is a boundary condition 0 such that

$$\big(\mathcal{B}_{\mathrm{op}(0)}, \mathcal{H}_{\mathrm{op}(0,0)}\big)$$

is a local conformal net

$$(\mathcal{A}, \mathcal{H}_0).$$

This means $(\mathcal{A}, \mathcal{H}_0)$ is a non-local net satisfying the following conditions:

1. single-valuedness on S^1;
2. locality:
$$I \cap J = \emptyset \quad \Rightarrow \quad [\mathcal{A}(I), \mathcal{A}(J)] = 0;$$
3. a vacuum vector $\Omega \in \mathcal{H}_0$ fixed by the Mobius group;
4. and some other properties.

The Local Conformal Net $(\mathcal{A}, \mathcal{H}_0)$

The pair $(\mathcal{A}, \mathcal{H}_0)$ is a non-local net satisfying the following extra conditions.

Single-valuedness

If $\widetilde{I}_1, \widetilde{I}_2 \in \widetilde{\mathcal{J}}$ project to the same interval $I \in \mathcal{J}$, then

$$\mathcal{A}(\widetilde{I}_1) = \mathcal{A}(\widetilde{I}_2).$$

Thus we write

$$I \in \mathcal{J} \longmapsto \mathcal{A}(I) \subset \mathfrak{L}(\mathcal{H}_0).$$

Locality and vacuum

$$I \cap J = \emptyset \quad \Rightarrow \quad [\mathcal{A}(I), \mathcal{A}(J)] = 0.$$

There is a vacuum vector $\Omega \in \mathcal{H}_0$ fixed by the action of

$$\mathrm{Mob} \subset \mathscr{G},$$

the subgroup of linear fractional transformations preserving S^1.

The Cardy-type CFT

Now restrict to the Cardy-type CFT. Set

$$\mathcal{H}_i := \mathcal{H}_{\mathrm{op}(i,0)}.$$

Then $\mathcal{H}_i$ is a right $\mathcal{A}$-module.

Open-string state space

Since one expects

$$\mathcal{H}_{\mathrm{op}(i,j)} = \mathcal{H}_{\mathrm{op}(i,0)} \boxtimes_{\mathcal{B}_{\mathrm{op}(0)}} \mathcal{H}_{\mathrm{op}(0,j)} = \mathcal{H}_{\mathrm{op}(i,0)} \boxtimes_{\mathcal{B}_{\mathrm{op}(0)}} \overline{\mathcal{H}_{\mathrm{op}(j,0)}},$$

we have

$$\mathcal{H}_{\mathrm{op}(i,j)} = \mathcal{H}_i \boxtimes_{\mathcal{A}} \overline{\mathcal{H}_j}.$$

Boundary Conditions Labeled by $\mathcal{A}$-Modules

The boundary conditions in the Cardy case are labeled by right $\mathcal{A}$-modules, equivalently solitonic $\mathcal{A}$-modules.

Local modules

For simplicity, in this talk we restrict to local $\mathcal{A}$-modules:

$$\text{left action} = \text{right action}.$$

We simply call them $\mathcal{A}$-modules.

Example

If $\mathcal{A}$ is the Virasoro conformal net, then $\mathcal{A}$-modules correspond to unitary representations of the Virasoro algebra.

Part III

Bulk Fields and Closed String State Space

Closed CFT

For each local $\mathcal{A}$-module $\mathcal{H}_i$, there is a canonical σ-twisted $\mathcal{A}^{\otimes 2}$-module structure on the same Hilbert space, where

$$\sigma \in \mathrm{Aut}(\mathcal{A}^{\otimes 2})$$

is the permutation of the two tensor factors.

Longo–Xu

We denote this twisted module by

$$\mathcal{H}_{\sqrt{i}}.$$

This is the Longo–Xu result (04); a vertex-algebra version appeared in Barron–Dong–Mason (02).

The State Space for the Closed CFT

The state space for the closed CFT is

$$\mathcal{H}_{\mathrm{cl}} = \mathcal{H}_{\sqrt{0}} \boxtimes_{\mathcal{A}^{\otimes 2}} \overline{\mathcal{H}_{\sqrt{0}}},$$

as an untwisted $\mathcal{A}^{\otimes 2}$-module.

Bulk net

We have a net of von Neumann algebras

$$\mathcal{B}_{\mathrm{cl}} : \ O \longmapsto \mathcal{B}_{\mathrm{cl}}(O) \subset \mathfrak{L}(\mathcal{H}_{\mathrm{cl}}),$$

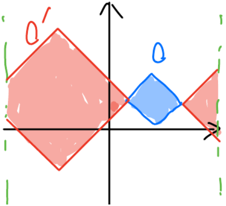
where O is a double cone in $\mathbb{R}^{1,1}_{\mathrm{cl}}$.

$\mathbb{R}^{1,1}_{cl}$

The net satisfies properties such as

$\mathscr{G}_c \times \mathscr{G}_c$-covariance, locality, **bulk-bulk Haag duality**.

Bulk-Bulk Haag Duality



Single double cone

If O' is the spacelike complement of O, then

$$\mathcal{B}_{\mathrm{cl}}(O)' = \mathcal{B}_{\mathrm{cl}}(O').$$

Multi double-cones

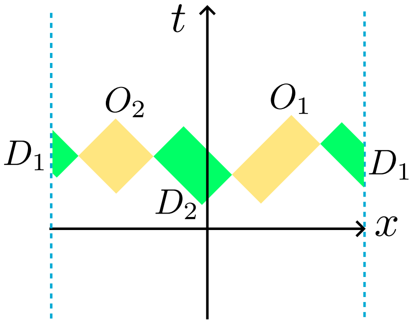
If $O_1, \ldots, O_n$ are mutually spacelike separated and

$$(O_1 \cup \cdots \cup O_n)' = D_1 \cup \cdots \cup D_n,$$

then

$$\big(\mathcal{B}_{\mathrm{cl}}(O_1) \vee \cdots \vee \mathcal{B}_{\mathrm{cl}}(O_n)\big)' = \mathcal{B}_{\mathrm{cl}}(D_1) \vee \cdots \vee \mathcal{B}_{\mathrm{cl}}(D_n).$$

Two and Multi Double-Cones



Two double-cones version

In the configuration shown on the left,

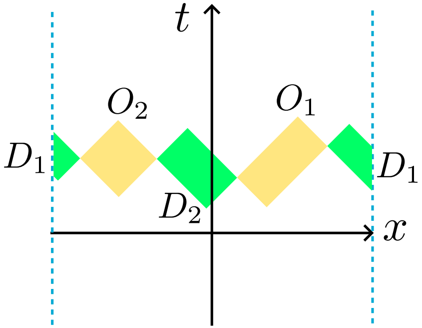
$$(O_1 \cup O_2)' = D_1 \cup D_2,$$

and

$$\big(\mathcal{B}_{\mathrm{cl}}(O_1) \vee \mathcal{B}_{\mathrm{cl}}(O_2)\big)' = \mathcal{B}_{\mathrm{cl}}(D_1) \vee \mathcal{B}_{\mathrm{cl}}(D_2).$$

The same statement holds for multi double-cones.

Modular Invariance



In Euclidean CFT, this version of Haag duality corresponds to **modular invariance**.

Multi double-cone version also holds.

Part IV

Bulk Fields Acting on Open String State Spaces

$\mathcal{B}_{\mathrm{cl}}$ Acts on $\mathcal{H}_{\mathrm{op}(i,j)}$

The closed net restricts to the open-string state space

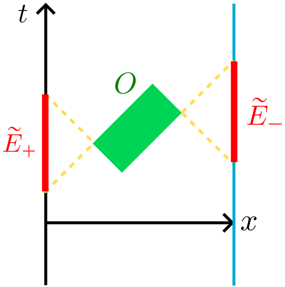
$$\mathcal{H}_{\mathrm{op}(i,j)} = \mathcal{H}_i \boxtimes_{\mathcal{A}} \overline{\mathcal{H}_j}.$$

Thus, for each double cone $O \subset \mathrm{Int}\mathbb{R}^{1,1}_{\mathrm{op}} = (0,\pi) \times \mathbb{R}$, we get an action

$$\mathcal{B}_{\mathrm{cl}}(O) \curvearrowright \mathcal{H}_{\mathrm{op}(i,j)}.$$

The action satisfies, among other things, the **bulk-boundary Haag duality**.

Bulk-Boundary Haag Duality



Let $\widetilde{E}_+$ and $\widetilde{E}_-$ be the two boundary intervals in the spacelike complement of the bulk double cone O, as in the picture.

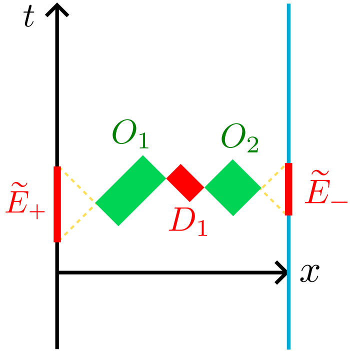
Bulk-boundary Haag duality

$$\mathcal{B}_{\mathrm{cl}}(O) \quad \text{and} \quad \mathcal{B}_{\mathrm{op}(i)}(\widetilde{E}_+) \vee \mathcal{B}_{\mathrm{op}(j)}(\widetilde{E}_-)$$

are commutants of each other.

This corresponds to the **Cardy consistency condition** in Euclidean CFT.

Multi Double-Cones in the Strip



The multi double-cone version holds.

Theorem

Theorem

For any conformal net $\mathcal{A}$, there exists the open/closed CFT satisfying the previously mentioned properties, whose boundary conditions are labeled by local $\mathcal{A}$-modules.

Expectations. We expect the construction to generalize to:

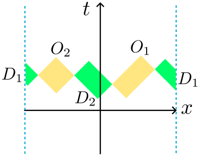
- non-local $\mathcal{A}$-modules;
- or even general non-local nets.

We expect that the open/closed CFT for $\mathcal{A} =$ Virasoro conformal net of $c \geq 25$ is the Minkowskian open/closed Liouville CFT.

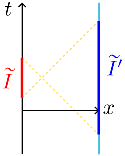
More generally, Toda CFTs are expected to be Cardy-type CFTs associated with principal $\mathcal{W}(\mathfrak{g})$ conformal nets at Toda central charge.

Three Haag Dualities

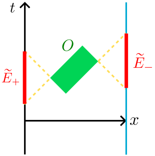
Haag duality	**Spacetime**	**Interpretation**
Bulk-bulk	$\mathbb{R}^{1,1}_{\mathrm{cl}}$	Modular invariance in Euclidean spacetimes
Boundary-boundary	$\mathbb{R}^{1,1}_{\mathrm{op}}$	Morita equivalence of boundary nets
Bulk-boundary	$\mathbb{R}^{1,1}_{\mathrm{op}}$	Cardy consistency condition in Euclidean spacetimes



Bulk-bulk



Boundary-boundary



Bulk-boundary